SRS/SDD Self-Review Report – Review 2

1. Run Information

Project code	FA26SE238
Project name	DroneOps Intelligence
Team	GFA26SE167
Supervisor	Pham Minh Tri; Dang Ngoc Minh Duc
AI tool and model	OpenAI Codex (repository-assisted review)
Run date	27 October 2026
Repository tag / commit	No review-2 tag present; commit 48aea3ce17b017bfbcf95ad34791fb53beaffab1
Registration form version used	Approved registration dated 05 September 2026 plus proposed BM03 changes
Version	Review 2 submission v1.0
Date	27 October 2026

2. Standard Prompt

Use this prompt without modification.

```
You are reviewing the SRS and SDD of a software capstone project against its
registration form and source code. Report only; do not change any file.

Inputs: the registration form, SRS, SDD (with diagrams), source code and
database script in this repository.

Check:
1. Scope: FRs in the SRS not covered by the registration form, and
   registration items with no FR.
2. Requirements: FRs without an actor or testable acceptance criteria;
   NFRs without metric, threshold and measurement method.
3. Data: entities missing from the Logical ERD; ERD entities or attributes
   missing from the database script, and tables not in the ERD.
4. States: for each main entity, states used in code but missing from its
   state machine; transitions with no triggering FR; states with no exit.
5. Flows: main business flows in activity diagrams with no implementation
   or no test.
6. Architecture: components or third-party services in code but not in the
   architecture diagram, and the reverse.
7. Design vs code: modules, APIs or business rules implemented differently
   from the SDD.

Output one table: ID | Category | Location | Finding | Severity
(High/Medium/Low) | Evidence. Then list the 5 most important findings.
If something cannot be verified from the repository, say so; do not guess.
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3. Findings and Team Response

Team response: Fixed (link to commit) · Kept (reason) · Planned (owner and due date). Do not delete any finding.

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-01	*Scope*	*Registration §3.2(b–g); SRS FR-19/NFR-M02; source*	*Registered Top-k similarity/vector retrieval is specified in the SRS but has no implementation in the source; the implemented OpenAI adapter is not a substitute for retrieved historical evidence.*	*High*	*Registration requires k-NN/vector search; SRS FR-19.1/BR-18/NFR-M02 retain it; source contains OpenAIService but no vector index or retrieval service.*	*Planned – Giang Nguyen Gia Khanh – 30 Nov 2026; implement and evaluate retrieval unless BM03 change is accepted.*

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-02	Scope	Report 1 §1.1/§7; SRS mobile statements; source	The project material identifies a Mobile App/operator mobile client, but the repository contains only the Next.js Web client and no defined mobile feature subset or mobile package.	High	Report 1 says "Web, Mobile App"; fe/ is Next.js; no Android/iOS/Flutter/React Native project was found.	Planned – Le Quoc Thong – 30 Nov 2026; remove mobile scope only if BM03 change is accepted, otherwise define and deliver its subset.
SR-03	Data	SDD §2 ERD/entity table; Domain/AppDbContext	The logical ERD/entity catalogue omits implemented records including MissionUpdateRequest, MissionValidationRun and DroneConnection, so design-to-schema traceability is incomplete.	High	Domain entities and AppDbContext DbSets exist; SDD entity table lists 21 entities but not these records.	Planned – Ha Quang Nghia – 06 Nov 2026; update ERD and entity catalogue from the verified schema.
SR-04	Flows	SRS main workflows; Report 5; CI	The four main workflows have designed cases but no recorded staging end-to-end execution evidence or pass results.	High	The original dossier recorded 40 integration/system and 10 unit designs. The submitted test report now contains a five-case executed integration subset with raw logs; UT-05 and UT-06 conditions have service-regression evidence. Full staging workflow acceptance is a separate verification scope.	Planned – all workflow owners – 30 Nov 2026; automate and retain seeded staging evidence.
SR-05	States	SRS BR-13; state diagrams; application services	Mission status spelling is not canonical across code and documents (for example InProgress and In Progress), which can create undocumented state variants.	High	TelemetryIngestionService and PostFlightService accept multiple literal variants; BR-13 names "In Progress".	Planned – Ha Quang Nghia – 06 Nov 2026; introduce canonical state constants and align diagrams/migrations.
SR-06	Architecture	SDD §1 architecture; gateway/	The Python MAVLink gateway is present in the repository but is represented only generically as an integration gateway; its deployable boundary, authentication and failure behavior are not fully specified.	Medium	gateway/gateway.py and requirements.txt exist; SDD architecture component text describes an integration gateway without a concrete deployment contract.	Planned – Le Quoc Thong – 13 Nov 2026; add gateway deployment/interface detail.
SR-07	Data	SDD §2 ERD; migrations	The ERD uses conceptual GeofenceRevision and ReviewDecision records whose mapping to actual revision tables/fields and MissionApproval/MissionUpdateRequest storage is not explicit.	Medium	Restriction revision routes and mission request migration exist; SDD does not provide a one-to-one physical mapping.	Planned – Ha Quang Nghia – 13 Nov 2026; add conceptual-to-physical mapping.
SR-08	States	SRS BR-15; incident state diagram; IncidentLifecycleService	Automatic recovery can transition an incident directly from Open to Resolved, but this trigger/path is not clearly distinguished from Manager resolution in the state diagram.	Medium	IncidentLifecycleService.ResolveOpenByRuleAsync supports rule recovery; diagram documents the common Open/Acknowledged/Resolved lifecycle.	Planned – Le Quoc Thong – 06 Nov 2026; label automatic recovery transition and triggering FR/rule.
SR-09	Requirements	SRS FR-01..FR-22; test documentation	Functional requirements have actors and behavior, but acceptance criteria are not consistently expressed as independent pass/fail statements linked to executable tests.	Medium	SRS detailed tables provide flows/postconditions; Report 5 cases are pending and not all FR sub-items have direct executed evidence.	Planned – all workflow owners – 20 Nov 2026; complete FR-to-test traceability and pass/fail criteria.
SR-10	Architecture	SDD architecture; API configuration	Firebase authentication, weather data and email delivery used/configured by the	Medium	FirebaseAdmin dependency/auth code, weather query service and	Planned – Le Quoc Thong – 13 Nov 2026; update

ID	Category	Location	Finding	Severity	Evidence	Team response
			application are not all shown as explicit third-party services and trust boundaries in the architecture diagram.		*mail services are present; SDD component list mentions external AI/email but not every provider boundary.*	*architecture and data-flow notes.*
SR-11	*Design vs code*	*SDD §3.1; mission services*	*The SDD describes a dedicated MissionApprovalService/review transaction abstraction while current behavior is split across controllers, MissionApprovalService and MissionValidationService with JSON proposal payloads.*	*Medium*	*MissionApprovalService and MissionValidationService share request validation/review responsibilities; migration stores BasePlanJson/ProposedPlanJson.*	*Kept – current split is accepted for Iteration 1; Ha Quang Nghia will align SDD responsibility text by 06 Nov 2026.*
SR-12	*Data*	*Logical ERD; database scripts*	*A complete attribute-level comparison cannot be verified from the diagram images/entity summary because the logical ERD does not enumerate every physical column, constraint and index.*	*Low*	*SDD provides entity descriptions; migrations define additional operational columns, partial indexes and checks.*	*Kept – conceptual ERD intentionally omits physical detail; migrations remain the physical source of truth.*

4. Top 5 Findings (as reported by the AI)

#	Finding ID	Summary
1	SR-01	Registered Top-k historical similarity retrieval has no source implementation.
2	SR-02	Mobile scope is stated but no mobile client or defined subset exists.
3	SR-03	Logical ERD omits implemented workflow/validation/gateway records.
4	SR-04	Four main workflows lack recorded staging end-to-end pass evidence.
5	SR-05	Mission state literals are not canonical across design and code.

5. Summary

Severity	Total	Fixed	Kept	Planned
High	*5*	*0*	*0*	*5*
Medium	*6*	*0*	*1*	*5*
Low	*1*	*0*	*1*	*0*

Every High finding marked Kept or Planned must also appear in section 5 (Known Gaps) of the Design Brief.

Automatic flight preparation and workflow ownership

Current ownership: MF1 / WF-01 — NghiaHQ (Ha Quang Nghia), planning and preflight; MF2 / WF-02 — ThongLQ (Le Quoc Thong), realtime monitoring, incidents and flight preparation; MF3 / WF-03 — DucDDA (Dang Dao Anh Duc), post-flight; MF4 / WF-04 — KhanhGNG (Giang Nguyen Gia Khanh), AI assessment. ThongLQ remains team leader and dossier coordinator.

Implemented scope

AutomaticFlightRequested is optional and defaults to false. Changes follow the mission proposal and approval workflow. This release prepares readiness; dispatchEnabled is false. Neither confirmation nor gateway enrollment uploads a mission, arms motors, sends TAKEOFF/AUTO/MISSION_START, or changes the mission to InProgress.

Confirmation conditions

For an approved, opted-in mission with a schedule, the backend opens confirmation at T minus 5 minutes and closes it at the scheduled start. The authenticated operator must keep a visible browser tab online. Presence is sent every 10 seconds and expires after 30 seconds. The selected, unrevoked mission connection must match the active UAV and flight-controller UID, with gateway heartbeat within 30 seconds and telemetry within 15 seconds.

Readiness evidence

Confirmation reruns preflight and binds readiness to the exact plan hash and connection ID. Readiness requires a passing preflight for that hash from the last minute. Changed plan/connection, stale presence or missing telemetry invalidates the conditions; the operator must review and confirm again. Dismissal, a hidden/offline tab or a missed window is not flight consent.

Future execution scope

Autopilot execution remains a separate development phase: firmware/airframe capability detection, mission compilation and upload/read-back, command IDs/deadlines and replay protection, scheduled condition rechecks, local pre-arm/failsafe enforcement, and ACK plus observed flight-state verification. Validate ArduPilot/PX4 SITL, propeller-free bench operation and supervised flight before enabling dispatch. AI advice cannot authorize launch.

Readiness review follow-up

Readiness is implemented; aircraft dispatch is disabled. Existing review findings and dates remain historical. Additional review must check the preparation state/API against the schema, boundaries and UI; verify fresh scoped evidence and absence of motor/start commands. Diagram updates and physical-aircraft acceptance remain outstanding, without claiming new pass evidence.